

# MAM · Exponential and Logarithmic Functions — Patterns and Solution Approaches

Basis: 42 past exam questions containing exponentials/logarithms from 2016–2025, marking keys + exam reports (the 2016, 2018, 2022–2025 reports all list logarithm-related content as a weakness: logarithm laws 2016/2024/2025; logarithmic graphs 2018/2024; reading and interpreting logarithmic scales 2023; converting between exponential and logarithmic form 2022). Original questions are in [topics/04\\_Exp\\_Log.pdf](#); two questions from the 2016 sample paper are also cited (2016S-A10, 2016S-A13; original paper in [papers/2016Sample\\_\\*](#)). Review status: has undergone one round of independent review and was revised according to feedback (see [Review\\_log.md](#)).

## Toolbox

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- Definition:  $y = \log_a x \Leftrightarrow a^y = x$ . The first step in many questions is to convert between logarithmic and exponential form (the key often awards 1 mark for "converts to index form / uses the inverse relationship").
- Logarithm laws:  $\log(mn) = \log m + \log n$ ;  $\log(m/n) = \log m - \log n$ ;  $\log m^k = k \log m$ ;  $\log_a a = 1$ ;  $\log_a 1 = 0$ ; change of base  $\log_a b = \log_c b / \log_c a$ .
- Constants can also be written as logarithms:  $4 = 4 \ln e = \ln e^4$ ;  $2 = \log_{10} 100$ ;  $9 = \log_{10} 10^9$ . This is a key step in 2025F-Q7(a), 2020F-Q6(c), 2024A-Q15(c), 2019A-Q17(b)(iii).
- Calculus:  $(\ln f)' = f'/f$ ;  $\int f'/f \, dx = \ln|f| + c$  (in exams  $f(x) > 0$  generally, and the key often writes  $\ln f(x) + c$ );  $(e^{kx})' = ke^{kx}$ .

## Marking panel reminders

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- Solving  $2 \ln x = \dots$ :  $\ln x$  requires  $x > 0$ , so negative roots must be rejected (the 2025F-Q7 key explicitly requires only one solution).
- "How many times" questions on logarithmic scales: the difference in readings is not the answer; the difference must still be converted to a power of 10 (the 2016A-Q12 key awards 1 mark for "subtracts Richter magnitudes", then converts to exponential form). The 2023 report calls for stronger interpretation and reading of logarithmic scales (see 2023F-Q4(c): reading values from the graph of  $y = \log_{10} x$ ).
- When a question requires the use of a graph, show the reading process on the graph (2024A-Q15(a): You must show clearly how you have used the graph).
- Application questions require units. If asked for rate of change, give a signed value (e.g. the answer to 2018A-Q9(c) is  $-0.164 \text{ mg/L/h}$ ); only if asked for rate of decrease should you write it as a positive number and state that it is decreasing (2024A-Q8(c) key: "converts to a rate of decrease").

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## Pattern 1: Expressing Logarithms Using Given Letters (CF)

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**Recognition:** "Let  $p = \ln 2$ ,  $q = \ln 3$ ,  $r = \ln 5$ . Express  $\ln 6.25$  in terms of ..." " $\log_{10} 2 = x$ ,  $\log_{10} 7 = y$ ".

**Approach:** Factor the argument into products, quotients and powers of the given numbers:  $6.25 = 25/4 = 5^2/2^2 \Rightarrow 2r - 2p$ ;  $17.5 = 7 \times 10 / 4 \Rightarrow y + 1 - 2x$ ;  $14 = 2 \times 7$ . **Change-of-base type (2025F-Q7(b)):**  $m = \log_3 6$ ,  $n = \log_6 5$ , find  $\log_3 10$ . First  $\log_3 10 = \log_3 2 + \log_3 5$ ; then  $\log_3 2 = \log_3 6 - 1 = m - 1$ ,  $\log_3 5 = \log_3 6 \cdot \log_6 5 = mn \Rightarrow \log_3 10 = m - 1 + mn$ .  **$e^{\{p+q\}}$  type:**  $e^{\{\ln 2 + \ln 3\}} = e^{\{\ln 6\}} = 6$  (index laws plus inverse relationship).

**Marking points:** ✓ factorise the argument ✓ use the correct logarithm laws ✓ final expression.

**Past exam questions:** 2017F-Q7, 2023F-Q2(a)(b), 2024F-Q5(a)(b), 2025F-Q7(b).

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## Pattern 2: Solving Exponential / Logarithmic Equations (CF exact values)

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### Approach

- Collect like terms:  $4e^{2x} = 81 - 5e^{2x} \Rightarrow 9e^{2x} = 81 \Rightarrow e^{2x} = 9 \Rightarrow x = \ln 3$ .
- Quadratic in  $e^x$ : let  $u = e^x$ , factorise, and reject roots with  $u \leq 0$  (2020F-Q7(a):  $e^x(e^x - 4) = 0 \Rightarrow$  only  $x =$

ln 4).

- Logarithmic equations: first combine all terms into a single logarithm ( $\log_2(x+y) + 2 = \log_2(x-2y) \Rightarrow \log_2[4(x+y)] = \log_2(x-2y)$ ), then remove the logarithms, and finally **check the domain**.

Past exam questions: 2017F-Q3, 2016F-Q1(b), 2020F-Q7(a), 2024F-Q5(c), 2025F-Q7(a).

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### Pattern 3: Logarithmic Function Graphs and Transformations

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**Recognition:** "sketch  $y = \ln(x-2) + 1$ " "determine the equation of  $g$ " "explain why  $p = 5$ " "find  $m, q$ ".

**Approach**

- $y = m \log_a(x-p) + q$ : vertical asymptote  $x = p$ ; always passes through  $(p+1, q)$ ; increasing when  $a > 1$  and  $m > 0$ .
- Three essentials for sketching (1 mark each in the key): **asymptote** (draw and label its equation; a dashed line is conventional), **one key point** (e.g.  $(3, 1)$ ), **correct shape** (including behaviour near the asymptote).
- Determine parameters from a graph: first use the asymptote to find  $p$ , then substitute two points to solve for  $m$  and  $q$  (2021A-Q15).
- Recognising transformations:  $\ln(4x) = \ln x + \ln 4 \rightarrow$  translate up by  $\ln 4$ ;  $\ln \sqrt{x} = \frac{1}{2} \ln x \rightarrow$  vertical dilation by  $\frac{1}{2}$ ;  $\ln x + 4 = \ln(e^4 x) \rightarrow$  horizontal dilation by a factor of  $1/e^4$  (2020F-Q6).
- $\log_2(1/(x-1)) = -\log_2(x-1)$ : first reflect in the  $x$ -axis, then translate right by 1 (2022F-Q4(c)).

**Solving equations or estimating from a graph:** e.g.  $\sqrt{7} \Rightarrow \log_2 \sqrt{7} = \frac{1}{2} \log_2 7 \approx \frac{1}{2} \times 2.8 = 1.4 \Rightarrow$  read  $x \approx 2.6$  (2022F-Q4(a)(ii)).

Past exam questions: 2018A-Q8, 2019F-Q4, 2020F-Q6, 2021A-Q15, 2022F-Q4, 2024F-Q5(d).

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### Pattern 4: Exponential Growth/Decay Models ★ CA high-frequency

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**Recognition:**  $P = P_0 e^{kt}$ ,  $C = 4e^{-0.05t}$ ,  $T = T_0 - 175e^{-0.07t}$ , half-life, "in danger if below ...".

**Standard procedure**

- Initial value:** substitute  $t = 0$  (with units).
- Find  $k$ :** use the half-life  $\frac{1}{2} = e^{k \cdot 100.5}$ , or use two data points. Round to the required number of significant figures.
- Time to reach a value:** set the model equal to the target value and solve using  $\ln$ ; convert to a specific time according to the context, and **round as required by the question**. Example: in 2018A-Q9(d),  $4e^{-0.05t} = 2.5 \Rightarrow t = \ln 1.6 / 0.05 \approx 9.40 \text{ h} \Rightarrow 9 \text{ am} + 9 \text{ h } 24 \text{ min} = \mathbf{6:24 \text{ pm}}$  (for "latest", round down; the key notes 6:25 would be too late). State any roots outside the domain as rejected (2025A-Q9(b)).
- Rate of change:**  $dP/dt = kP$ , substitute the time; see the reminder above about signs.
- Piecewise models:** the initial value of the later model = the value at the end of the previous model (2017A-Q16(c), 2023A-Q6(e)).
- Long-term behaviour:** as  $t \rightarrow \infty$ ,  $e^{-kt} \rightarrow 0$ , so  $T \rightarrow T_0$  and rate of change  $\rightarrow 0$  (the three points in 2020A-Q15(e): rate of change  $\rightarrow 0$ , temperature tends to a constant, and the constant equals  $T_0$ ).

**Marking points:** ✓ set up the equation ✓ solve ✓ round or convert time as required ✓ differentiate ✓ substitute values.

Past exam questions: 2016A-Q9, 2016S-A10, 2017A-Q16, 2018A-Q9, 2020A-Q15, 2022A-Q14, 2023A-Q6, 2024A-Q8, 2025A-Q9(b).

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### Pattern 5: Logarithmic Scales (decibels, Richter, site ranking, moment magnitude)

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**Recognition:**  $L = 10 \log(I/I_0)$ ,  $M = \log_{10}(A/A_0)$ ,  $R = 2 \log_{10}(S/S_0)$ , ask "how many times larger / more intense".

**Approach (ratio questions)**

1. General form  $L = k \cdot \log_{10}(I/I_0)$ . Convert the two readings to exponential form:  $I_1 = I_0 \cdot 10^{\{L_1/k\}}$ ,  $I_2 = I_0 \cdot 10^{\{L_2/k\}}$ .
2. Divide;  $I_0$  cancels:  $I_2/I_1 = 10^{\{(L_2-L_1)/k\}}$ . **k varies by question**: decibels  $L = 10 \log \Rightarrow 90 \text{ dB vs } 60 \text{ dB}$  is  $10^3 = 1000$  times; sound pressure  $D = 20 \log (2016S-A13) \Rightarrow$  a 30 dB difference is  $10^{\{1.5\}} \approx 31.6$  times; Richter  $k = 1$  (2016A-Q12)  $\Rightarrow 10^{\{2.1\}} \approx 126$  times; site ranking  $R = 2 \log$  (2023F-Q4).
3. Find the reference value  $I_0$  or  $S_0$ : substitute one pair  $(L, I)$ , convert to exponential form and solve.

**Linearisation (log against t is a straight line)**:  $\log_{10}P = At + B \Rightarrow P = 10^B \cdot (10^A)^t$ ; the slope  $A$  reflects the growth rate (in 2019A-Q17(c), comparing growth rates at two temperatures requires referring to the slope or base).

**Marking points**: ✓ convert to exponential form ✓ write the ratio and cancel ✓ conclusion ("... times").

**Past exam questions**: 2016A-Q12, 2016S-A13, 2018A-Q18, 2019A-Q17, 2022A-Q14(c)–(e), 2023F-Q4, 2024A-Q15, 2025A-Q17.

## Pattern 6: Logarithmic models + differentiation + drawing a conclusion

**Recognise**:  $s = 40.6 \ln(0.15D)$ ,  $A(t) = b \log_4(t+1) + c$ ,  $P(x) = 20 \ln(x+a)/(x+5)$ .

**Approach**

- Determine parameters from given points:  $A(0) = 21 \Rightarrow c = 21$ ;  $A(1) = 53 \Rightarrow b \cdot \log_4 2 = 32 \Rightarrow b = 64$ .
- Interpret the derivative:  $ds/dD = 40.6/D$  decreases as  $D$  increases  $\Rightarrow$  for the same 8 m reduction, roads with a larger original stopping distance have a smaller reduction in speed limit (2025A-Q11(c)).
- Determine whether the target can be achieved: solve  $A(t) = 493$ , check whether  $t$  is reasonable (2024A-Q17(c)), and give a conclusion with reasons.
- Round according to context: round quantities down (2024A-Q17(b)); compare adjacent integers (2020A-Q13 compares 74 and 75).
- "during which week" :  $x$  is the week number from the start;  $1 < x < 2$  is in **week 2**, not week 1 (the key for 2021A-Q16(d) says "occurs in the second week" ).

**Past exam questions**: 2018F-Q6, 2020A-Q13, 2021A-Q16, 2024A-Q17, 2025A-Q11.

## Pattern 7: Calculus of $e^x$ / $\ln x$ (cross-reference)

The methods for this question type are covered in the differentiation and integration handouts; only the questions are listed here, to make it easy to practise by the "exponential and logarithmic" topic:

- Use  $d/dx(x \ln x)$  to evaluate  $\int \ln x$ : 2018F-Q7, 2023F-Q2(c)  $\rightarrow$  see [02\\_Integration\\_Methods.md](#) Pattern 2.
- Areas involving  $e^x$ ,  $\ln x$  (including inverse-function area), and the area correspondence between  $\int 1/x$  and  $\ln$ : 2017F-Q5, 2019F-Q5, 2020A-Q11, 2021F-Q7  $\rightarrow$  Integration handout Patterns 5, 6.
- Finding  $f$  from  $f'$ , and  $\int f'/f$ : 2020A-Q10  $\rightarrow$  Integration handout Pattern 1.
- Increment formula to estimate  $\ln 2.02$ : 2021F-Q3  $\rightarrow$  Differentiation handout Pattern 7.
- For functions involving  $\ln$ ,  $e^x$ , find stationary points and sketch the graph: 2016A-Q13, 2017A-Q14, 2020F-Q7(b)–(d), 2019A-Q12  $\rightarrow$  Differentiation handout Pattern 4.
- Derivative of  $a^x$  and the definition of  $e$ : 2018A-Q14.